

Introduction to Geophysics

Objective:

infer the subsurface (or whole Earth) structure

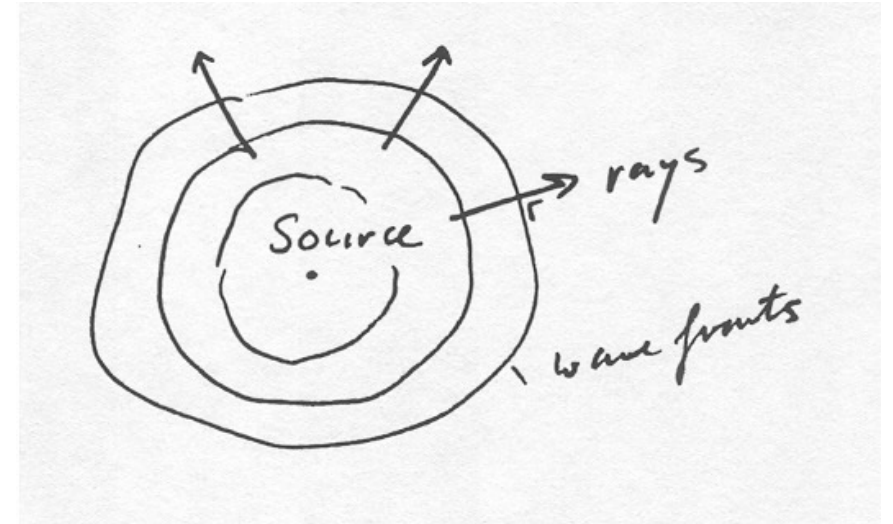
Methodologies:

- Seismology
- gravity

Wavefronts versus rays

A **wave front** is a surface connecting all points of equal travel time from the source.

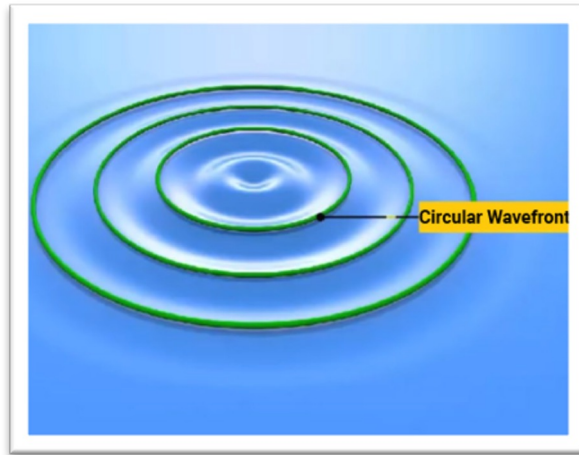
Rays are normal to the wavefront, and they point in the direction of wave propagation.



While the mathematical description of the wavefronts is rather complex, that of the rays is simple. For many applications it is convenient to consider rays rather than wavefronts.

Wavefronts versus rays

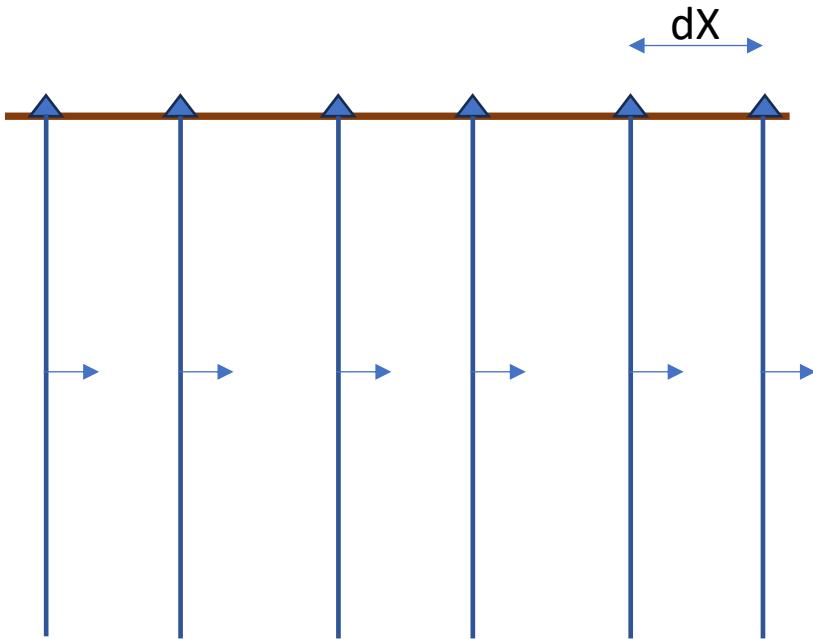
Circular wavefronts



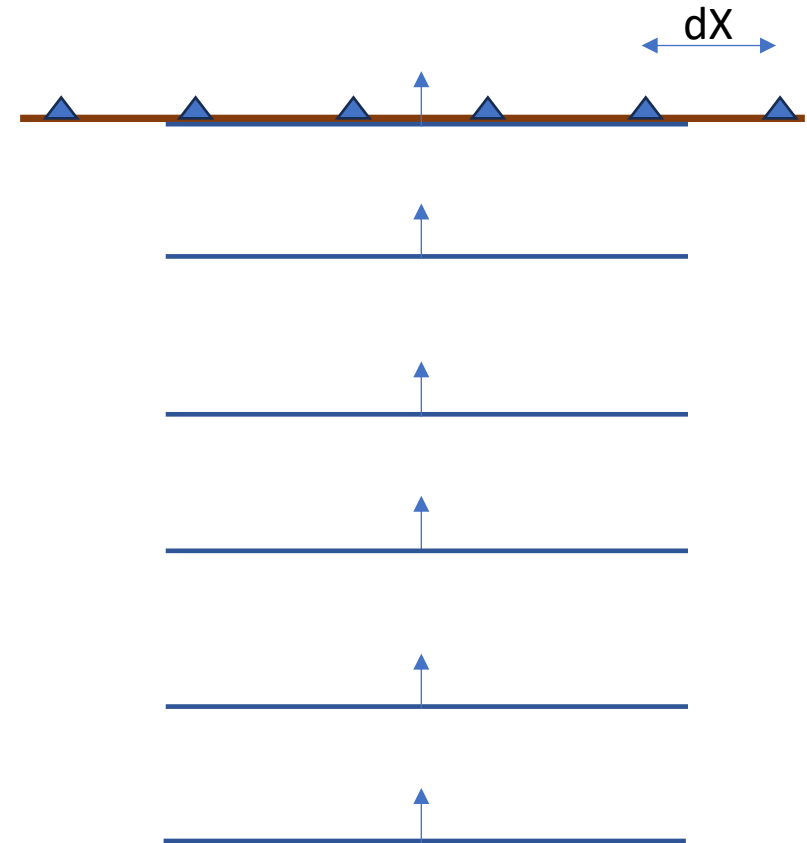
Planar wavefronts



Horizontal (or apparent) versus true velocities

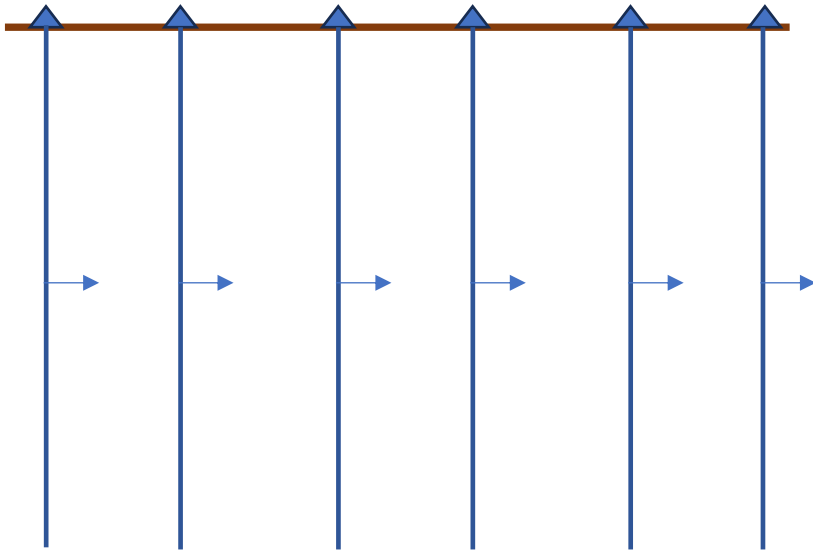


$dX/dt=?$

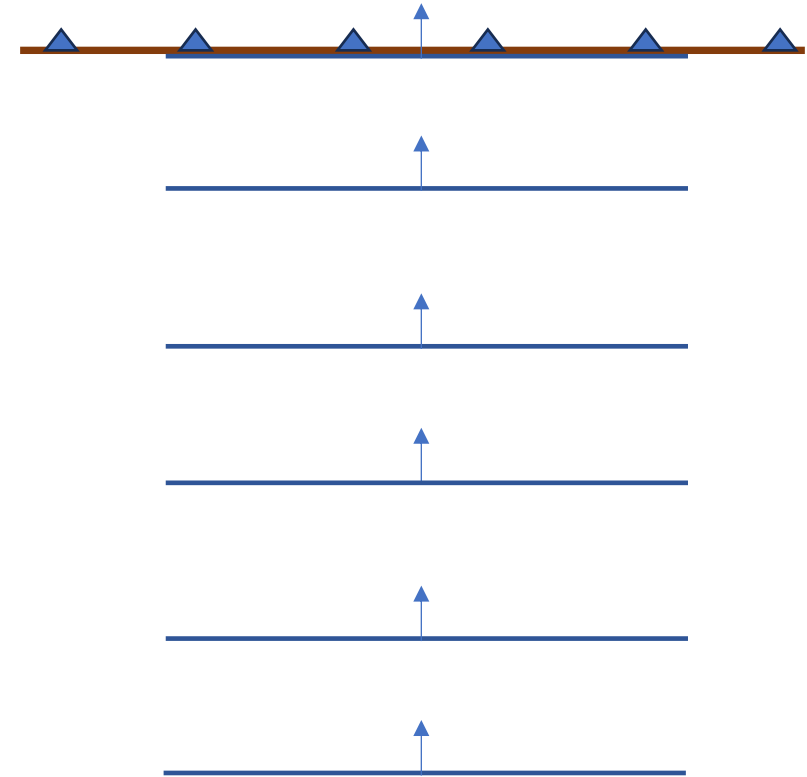


Horizontal (or apparent) versus true velocities

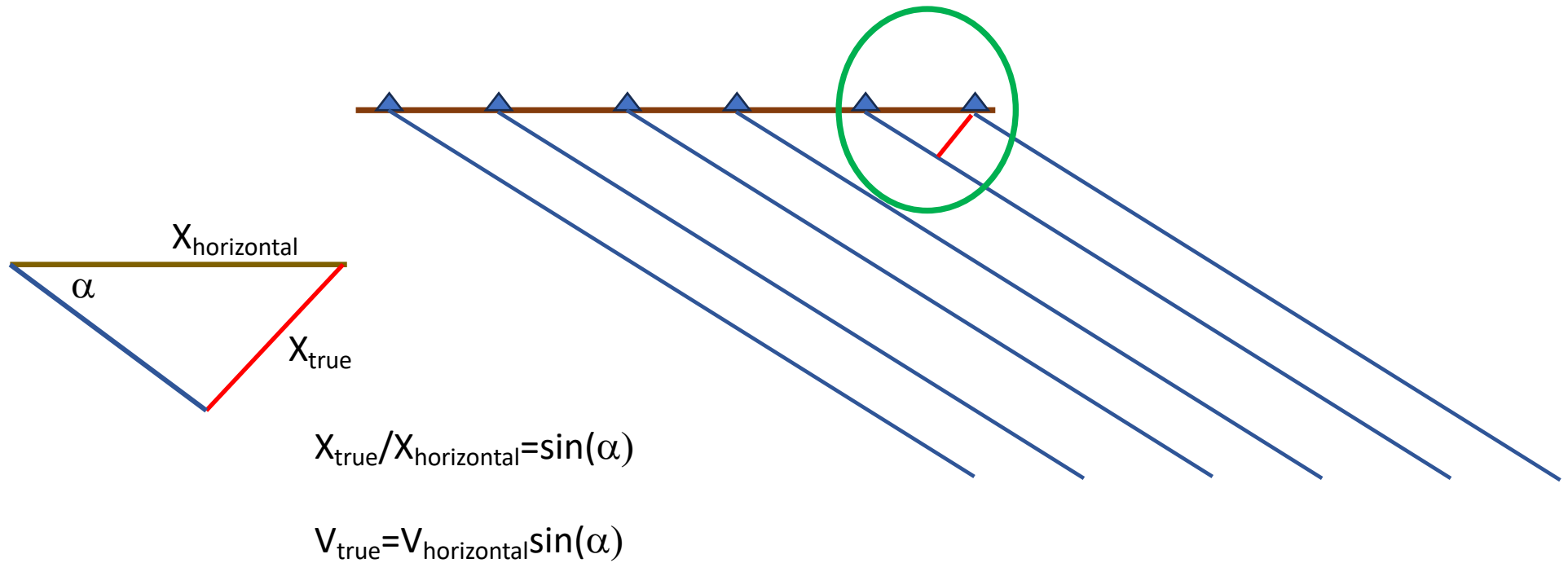
Horizontal = True



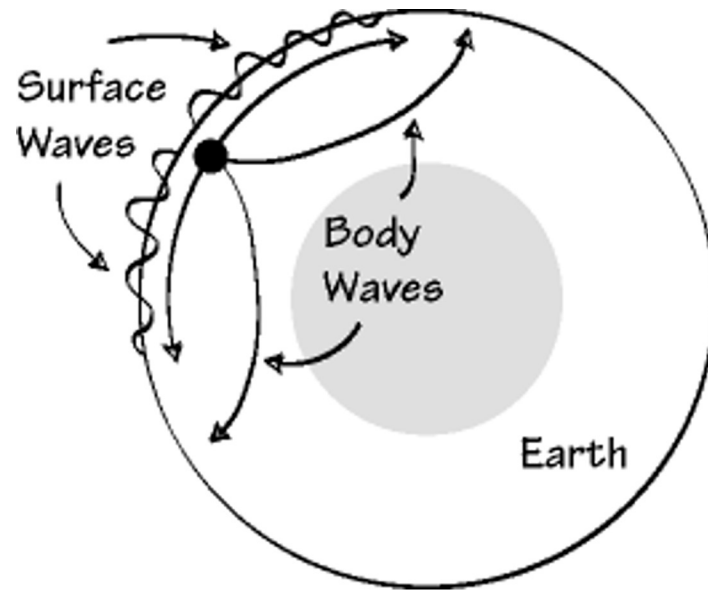
Horizontal $\rightarrow$ infinity



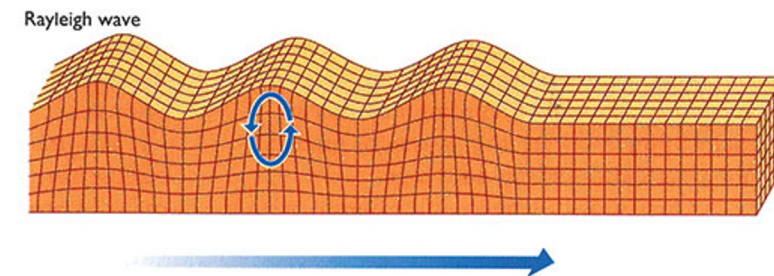
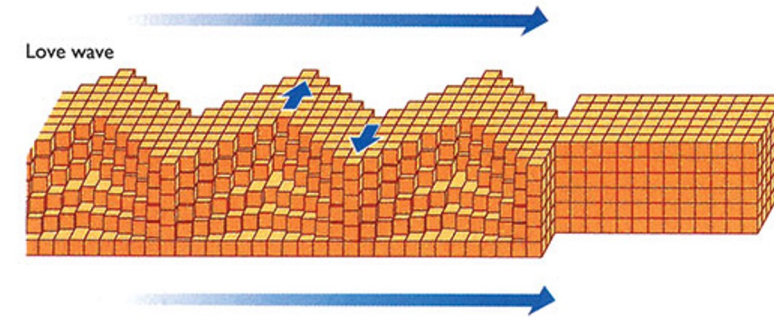
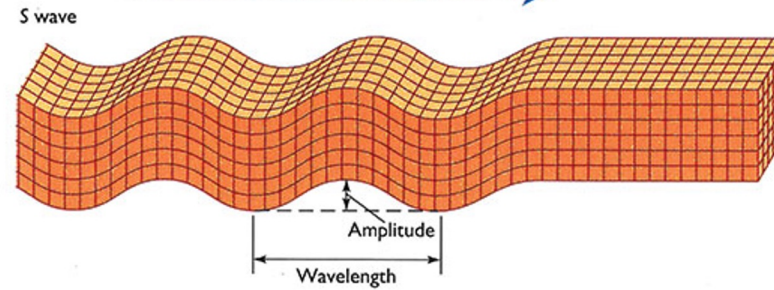
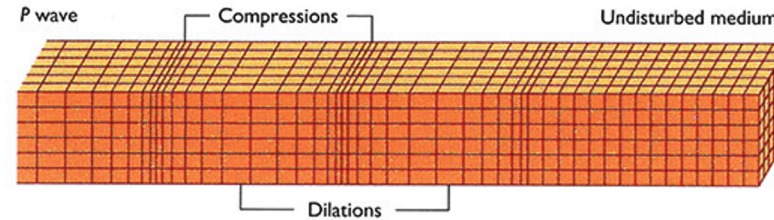
Apparent versus true velocities



Types of seismic waves

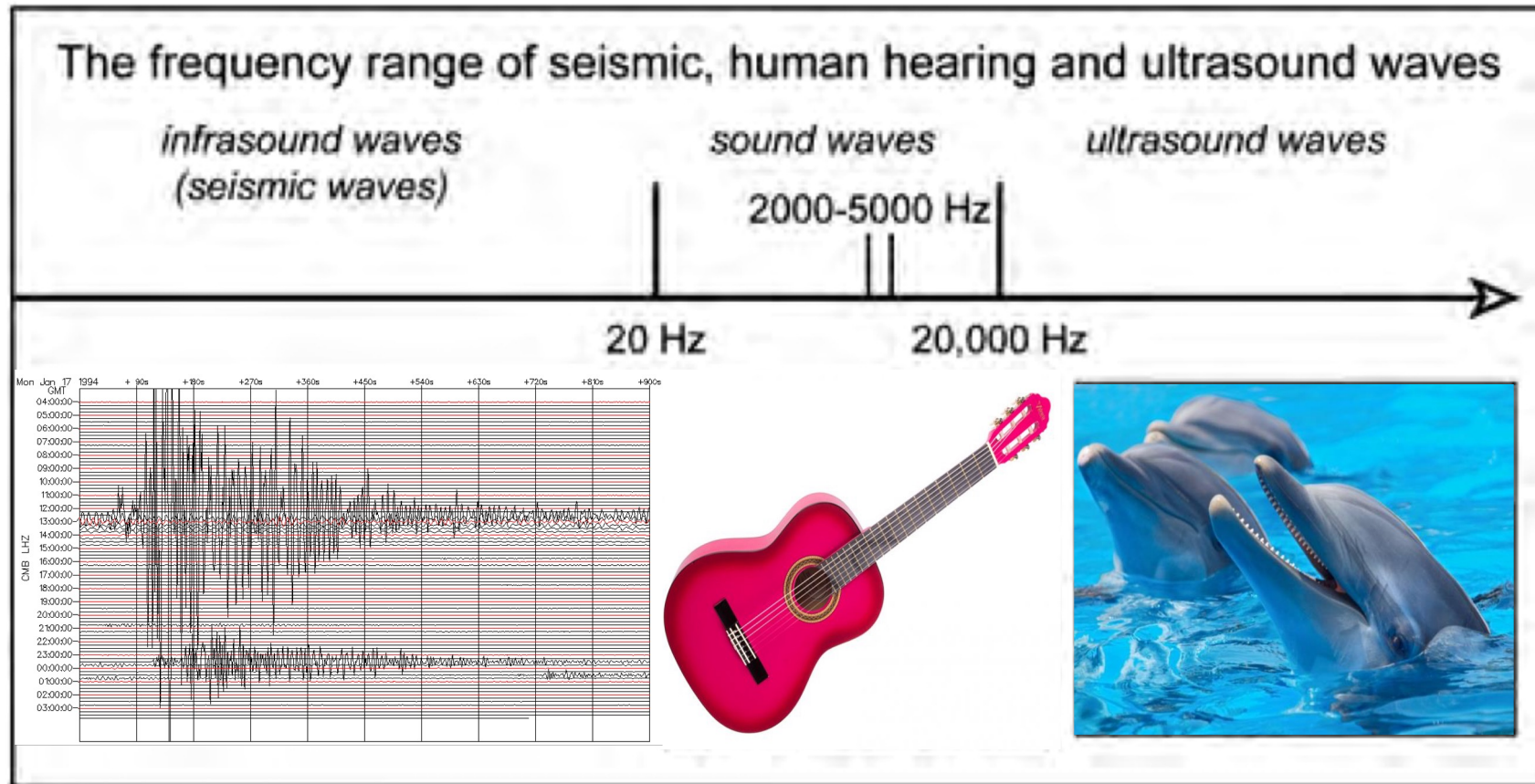


Body waves



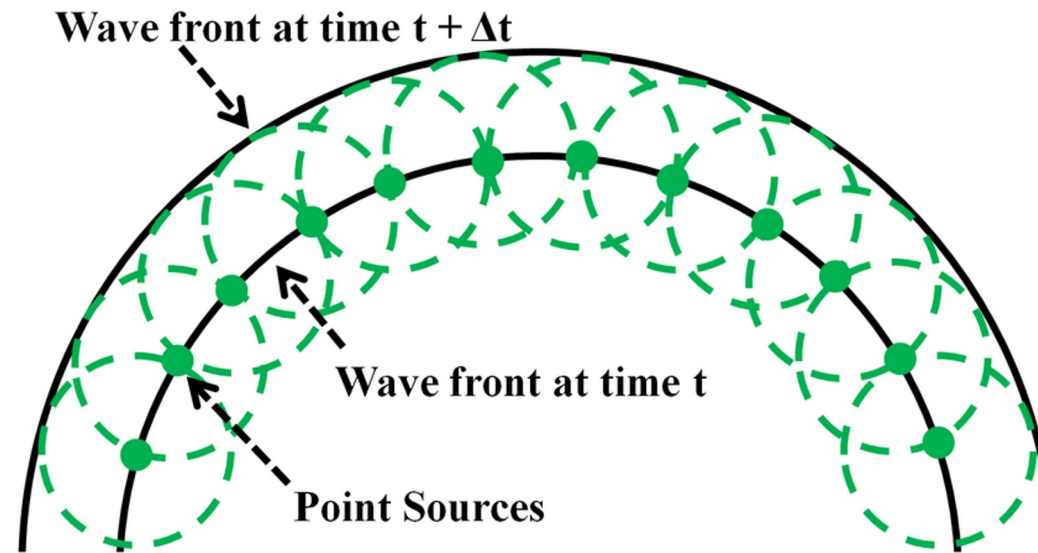
Surface waves

Wave spectrum



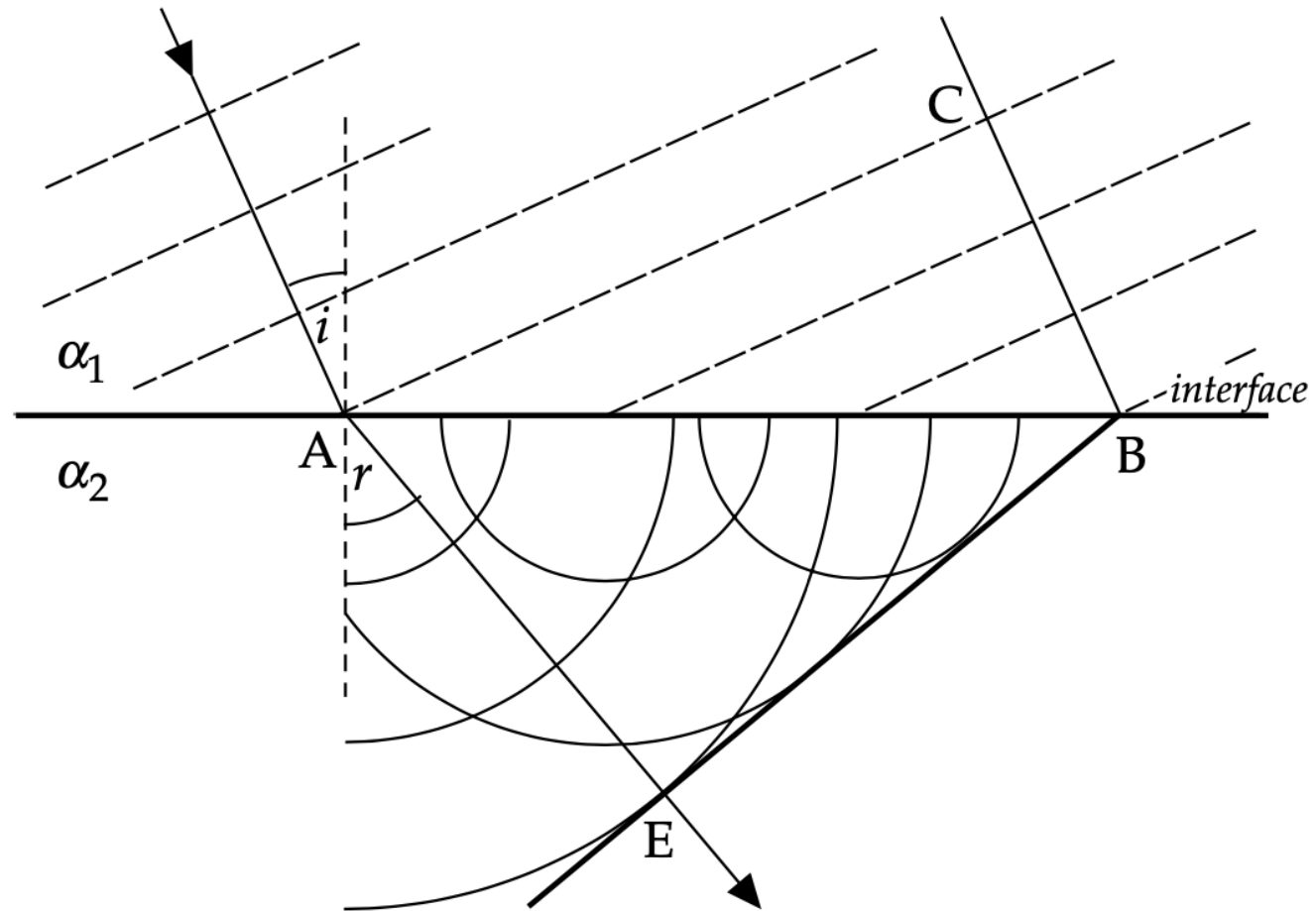
Huygen's principle

Every point on a wavefront is itself the source of spherical wavelets, and the secondary wavelets emanating from different points mutually interfere.

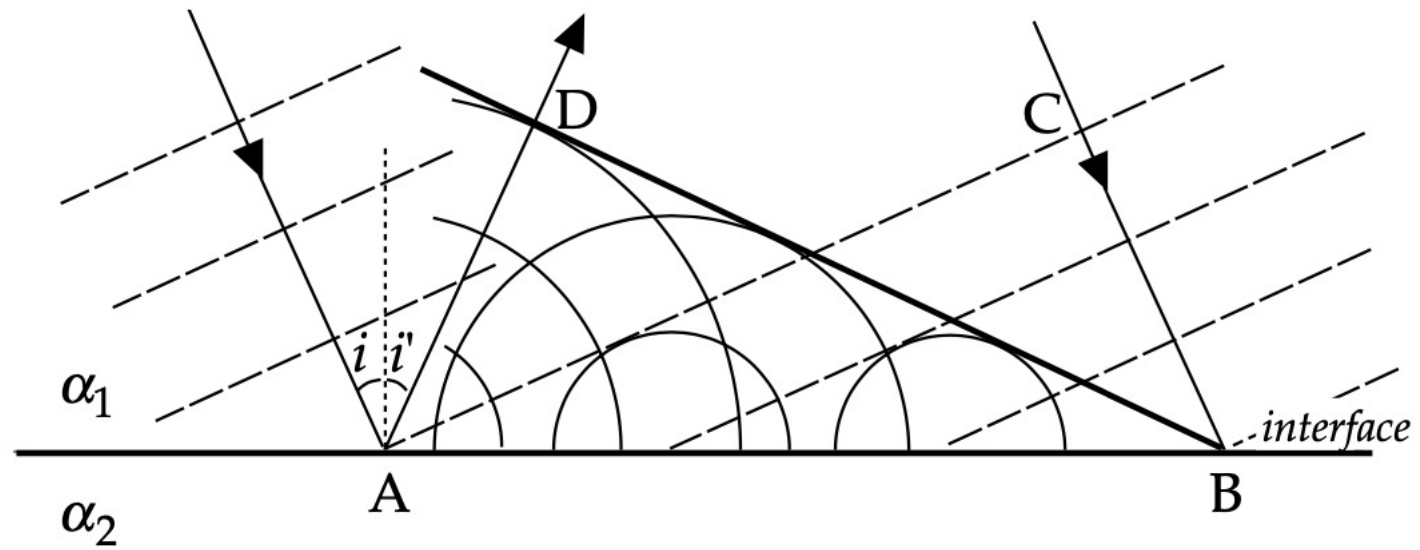


<https://www.youtube.com/watch?v=he8jSyRWvxE>

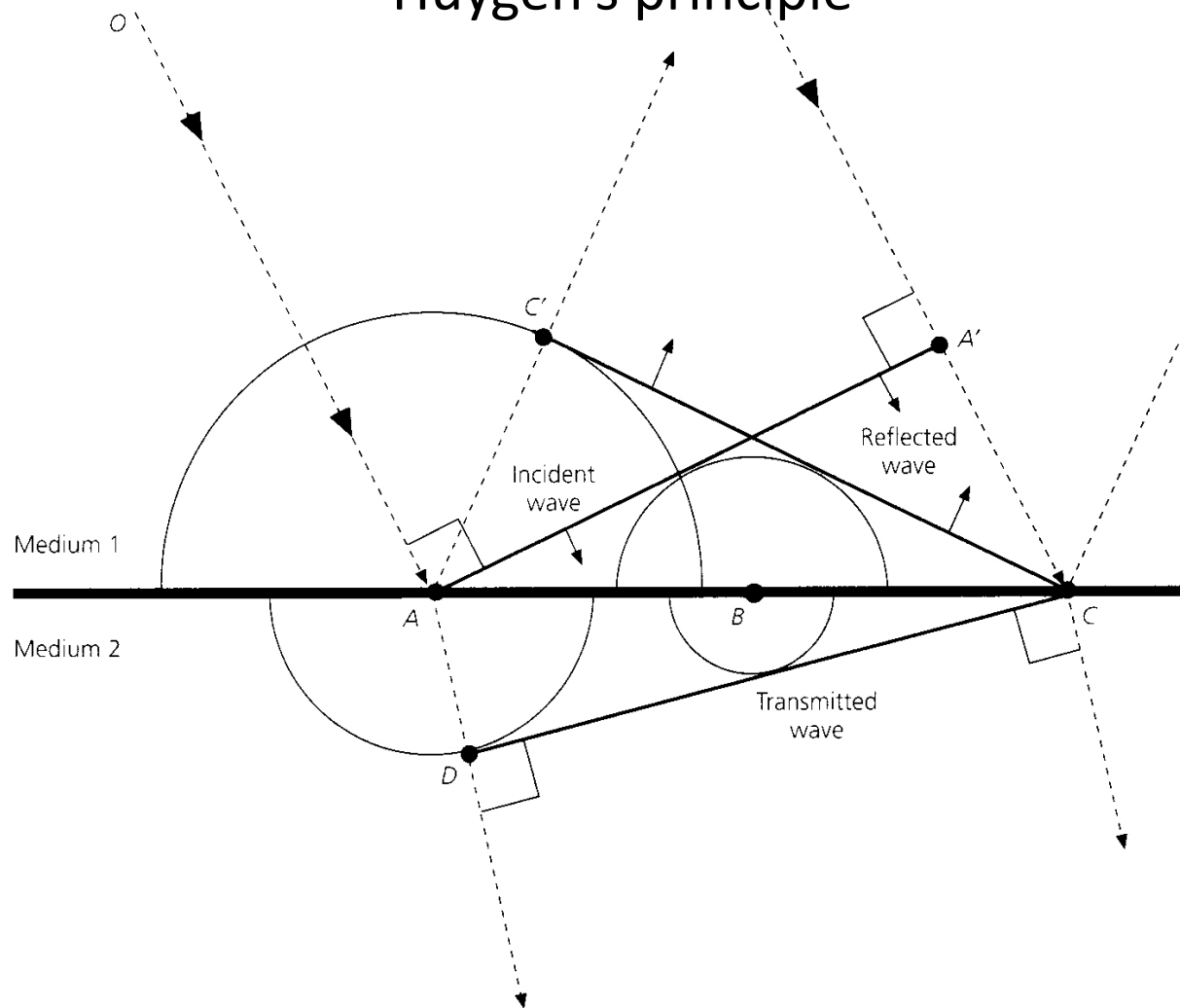
Huygen's principle



Huygen's principle

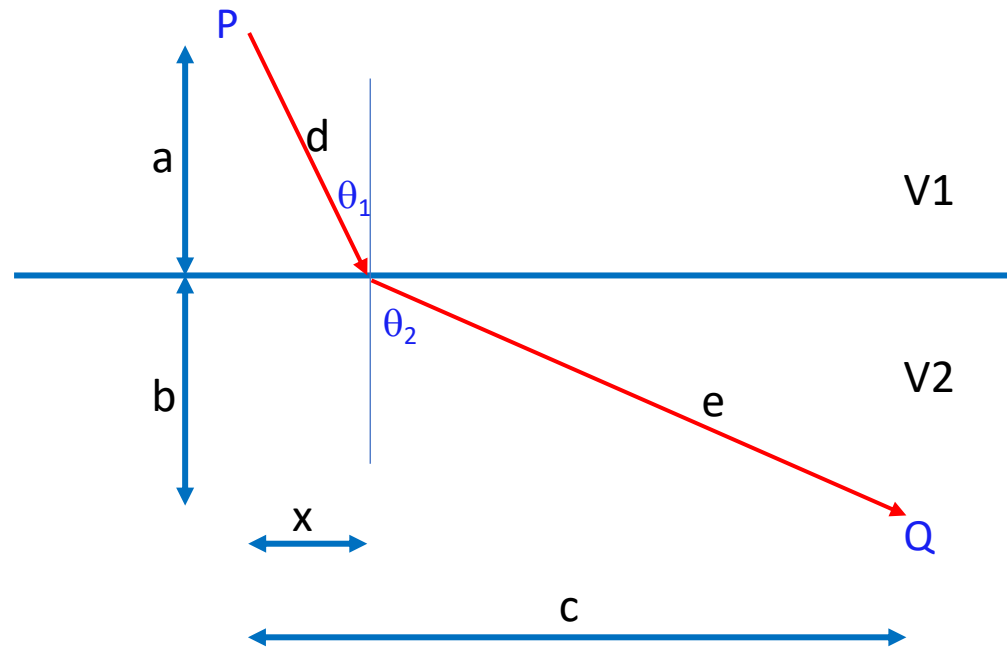


Huygen's principle



Fermat's principle and Snell's law

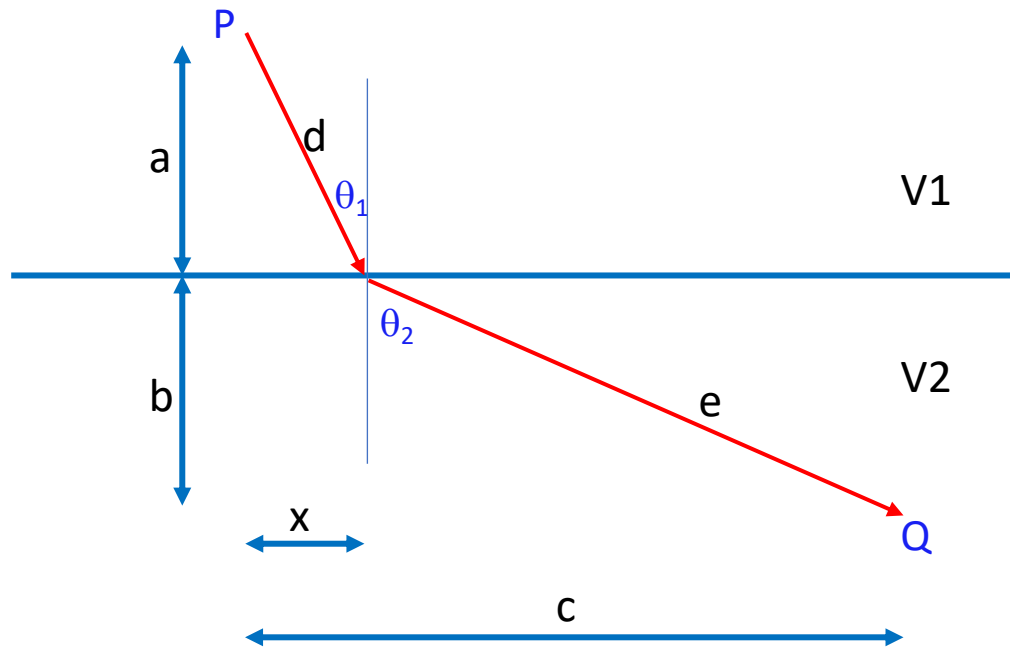
Fermat's principle states that, of the many possible paths between two points, the seismic ray follows the path that gives the fastest travel-time between the points.



What path will take a ray from point P to point Q?

Fermat's principle and Snell's law

Fermat's principle states that, of the many possible paths between two points, the seismic ray follows the path that gives the fastest travel-time between the points.



What path will take a ray from point P to point Q?

the travel-time from P to Q is

$$T_{P-Q} = \frac{d}{V_1} + \frac{e}{V_2} = \frac{\sqrt{a^2 + x^2}}{V_1} + \frac{\sqrt{b^2 + (c-x)^2}}{V_2}$$

since the wave speed in each layer is constant, the ray path is completely defined

by the position of P and Q, and the point where the ray crosses the interface.

Since this is a path for which the travel-time is minimum:

$$\frac{dT_{P-Q}}{dx} = 0 = \frac{x}{V_1 \sqrt{a^2 + x^2}} - \frac{c-x}{V_2 \sqrt{b^2 + (c-x)^2}}$$

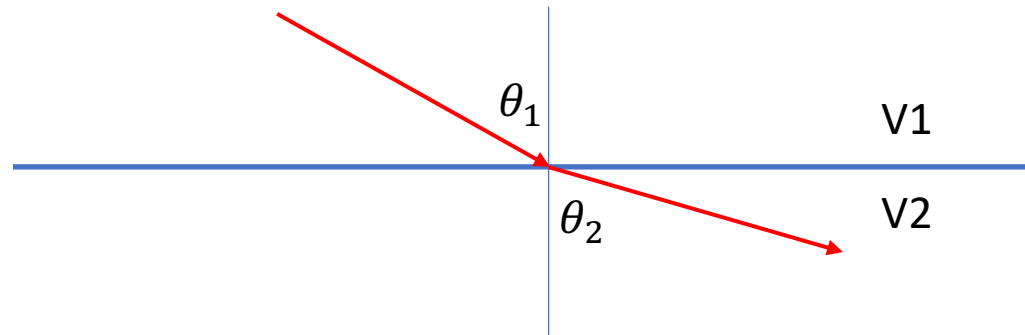
note that:

$$\frac{x}{\sqrt{a^2 + x^2}} = \sin \theta_1 \quad \text{and} \quad \frac{c-x}{\sqrt{b^2 + (c-x)^2}} = \sin \theta_2$$

Thus, we get: $\frac{\sin \theta_1}{V_1} = \frac{\sin \theta_2}{V_2}$

Snell's law and the ray parameter

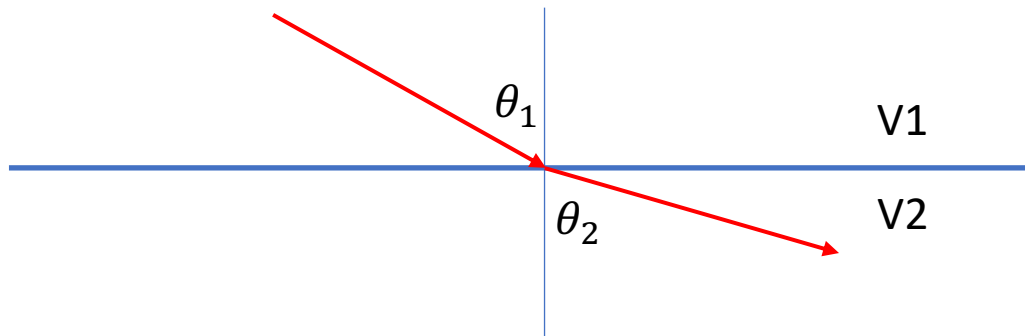
$$\frac{\sin(\theta_1)}{v_1} = \frac{\sin(\theta_2)}{v_2} \equiv P, \text{ referred to as the "ray parameter"}$$



Even in multilayered or highly heterogeneous media, the **ray parameter** remains constant along the ray path

Snell's law and the ray parameter

$$\frac{\sin(\theta_1)}{V_1} = \frac{\sin(\theta_2)}{V_2} \equiv P, \text{ referred to as the "ray parameter"}$$



Note that:

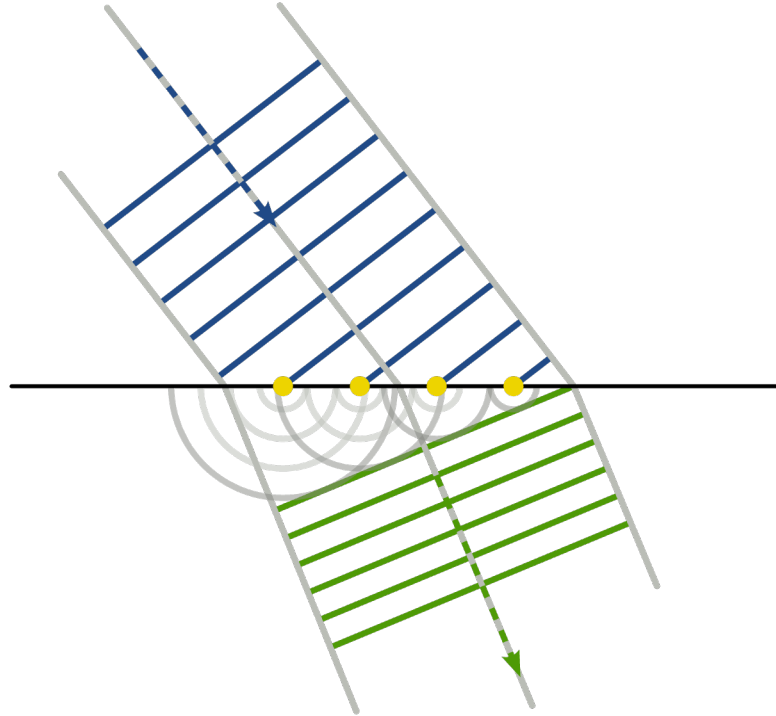
$$\frac{\sin(\theta)}{V} = \frac{1}{V_{horizontal}}$$

The ray parameter is the horizontal slowness

Even in multilayered or highly heterogeneous media, the **ray parameter** remains constant along the ray path

Snell's law and the ray parameter

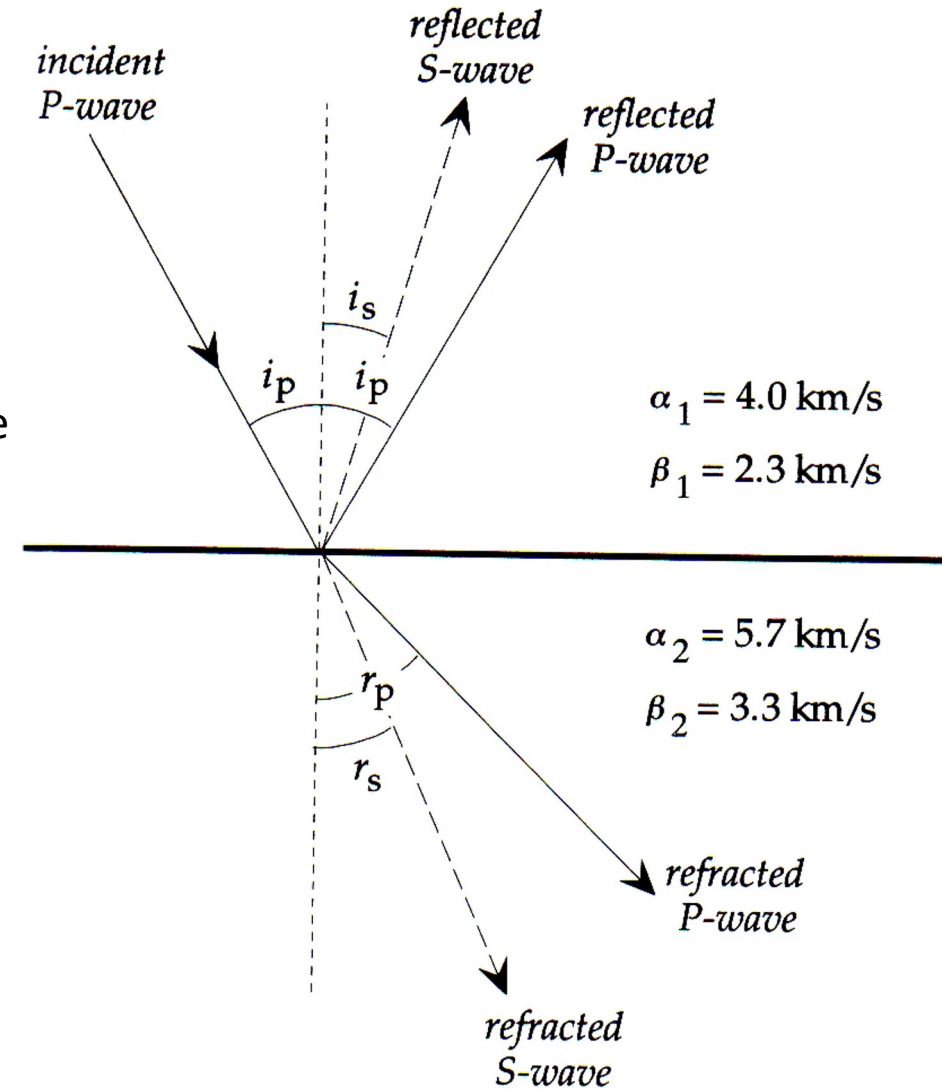
Here is how Snell's law may be explained in terms of Huygens principle



Snell's law for reflected and refracted P and S waves

The incidence angle of the reflected, transmitted and converted waves are controlled by an extended form of the Snell's law:

$$\frac{\sin(i_p)}{\alpha_1} = \frac{\sin(i_s)}{\beta_1} = \frac{\sin(r_p)}{\alpha_2} = \frac{\sin(r_s)}{\beta_2}$$

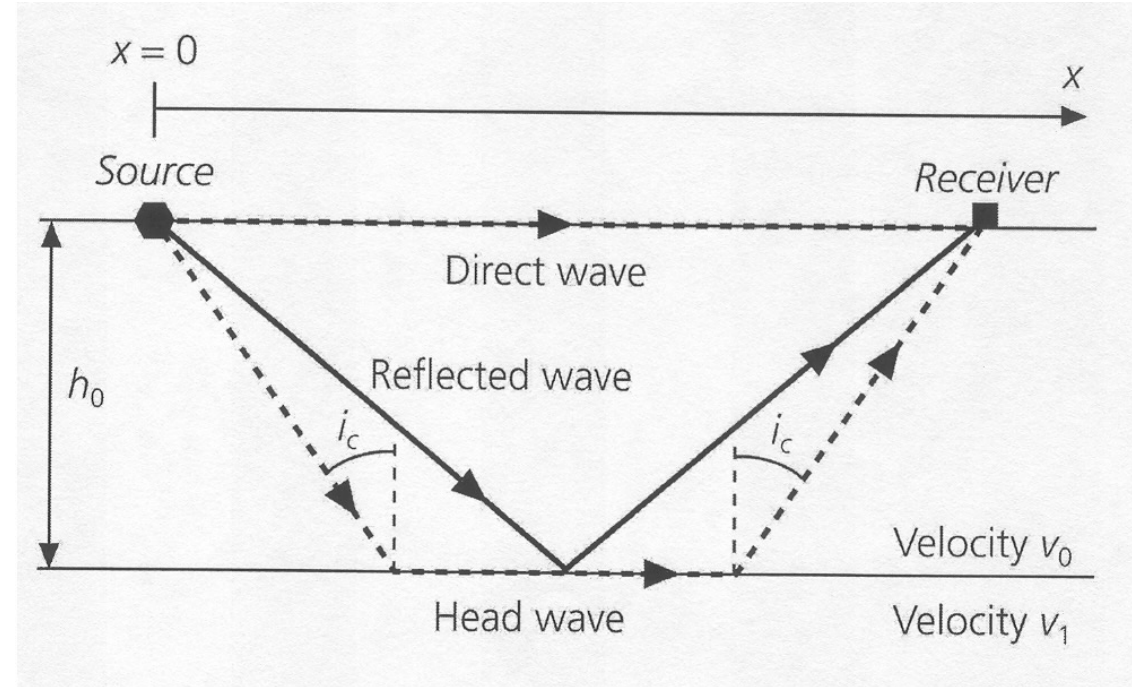


Head waves and the critical angle

Consider a seismic wave generated near the surface and recorded by a seismic station at some distance

In the simple case of a 2-layer medium, the following arrivals are expected:

- the arrival of the **direct wave**.
- the arrival of the **reflected wave**.
- the arrival of the **refracted wave**.

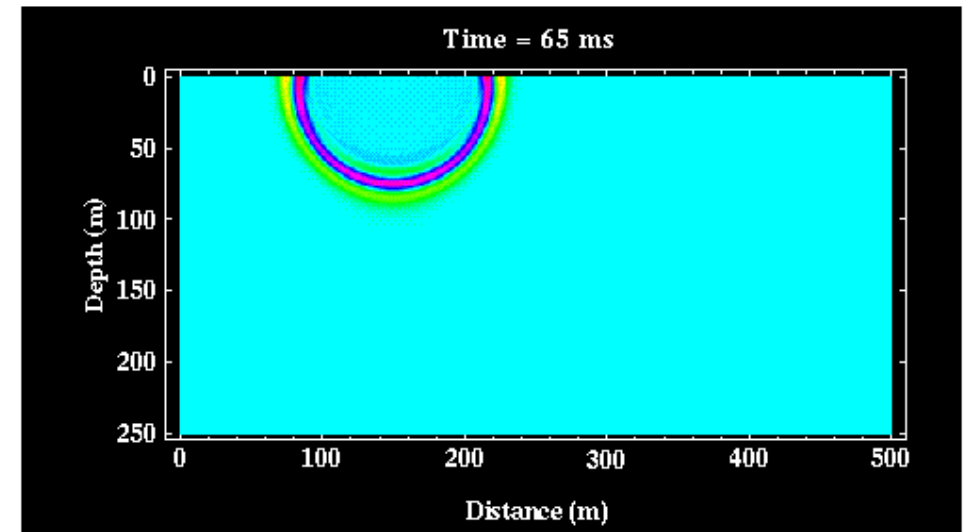
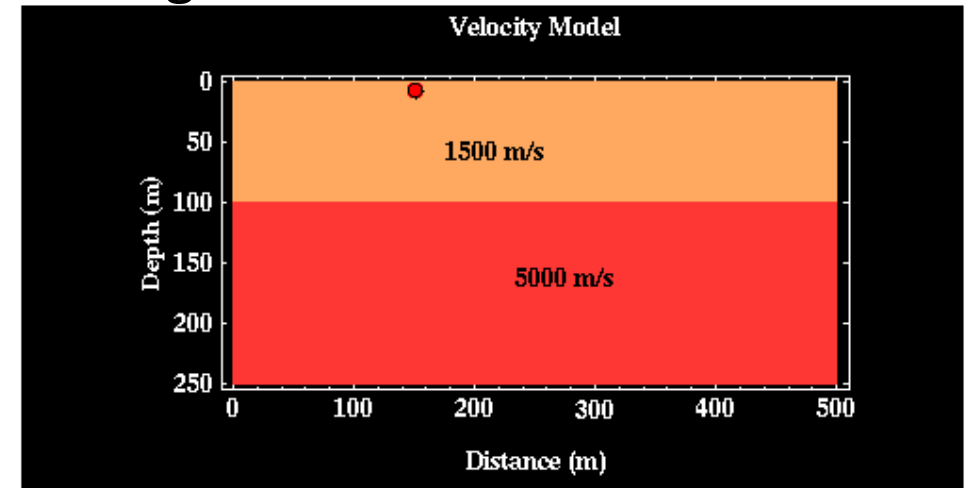


The critical angle is a special case of Snell's law where the refracted ray forms 90 degrees with the vertical:

$$\sin(i_c) = v_0/v_1$$

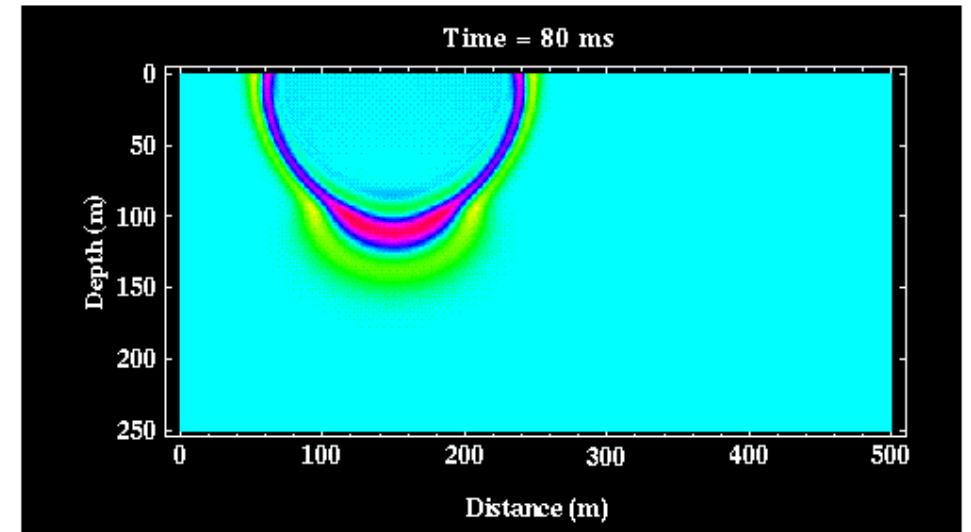
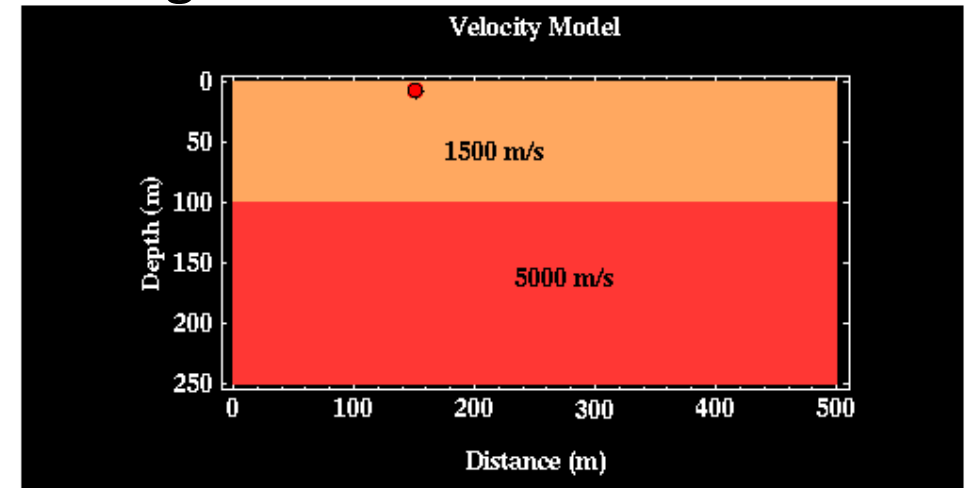
Head waves and the critical angle

Snapshots of numerically simulated
wave fronts: 65 ms



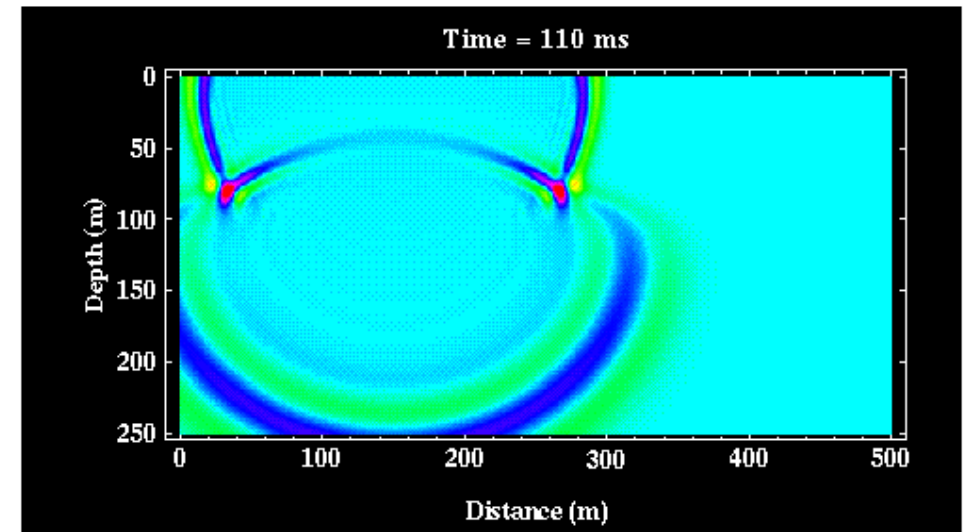
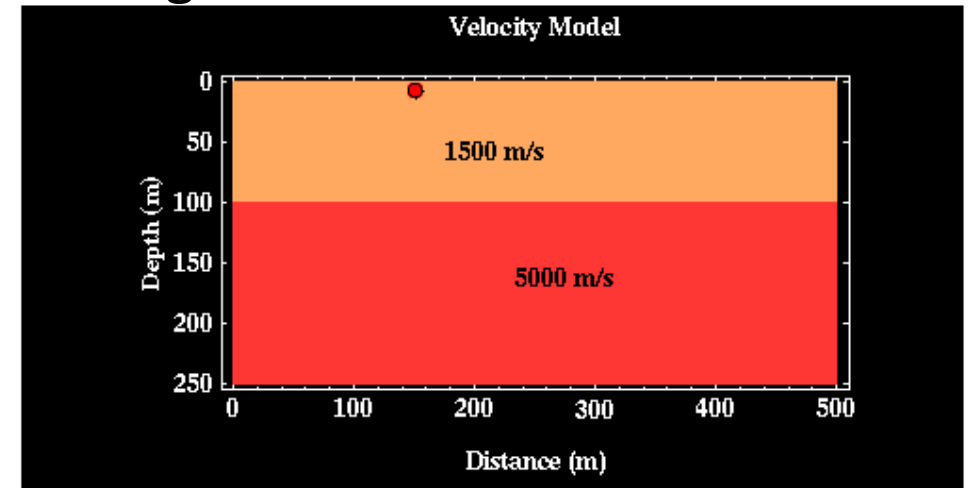
Head waves and the critical angle

Snapshots of numerically simulated
wave fronts: 80 ms



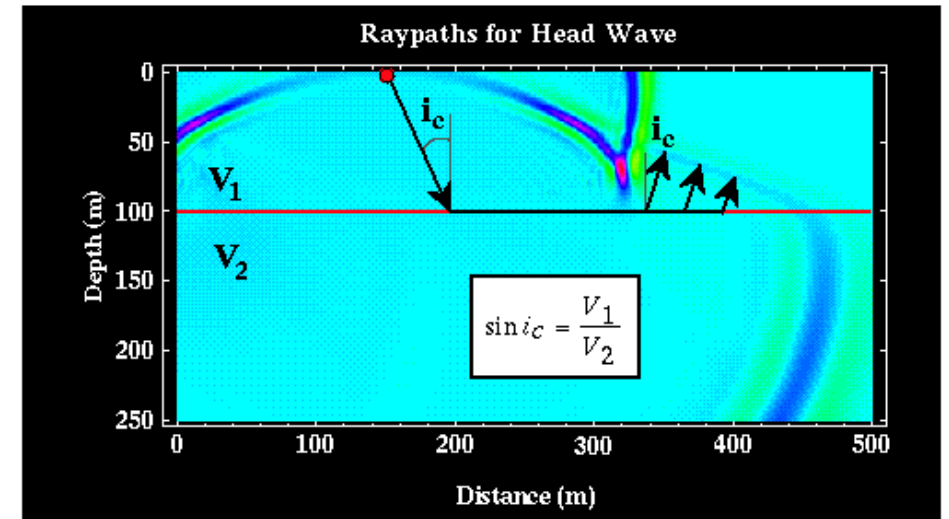
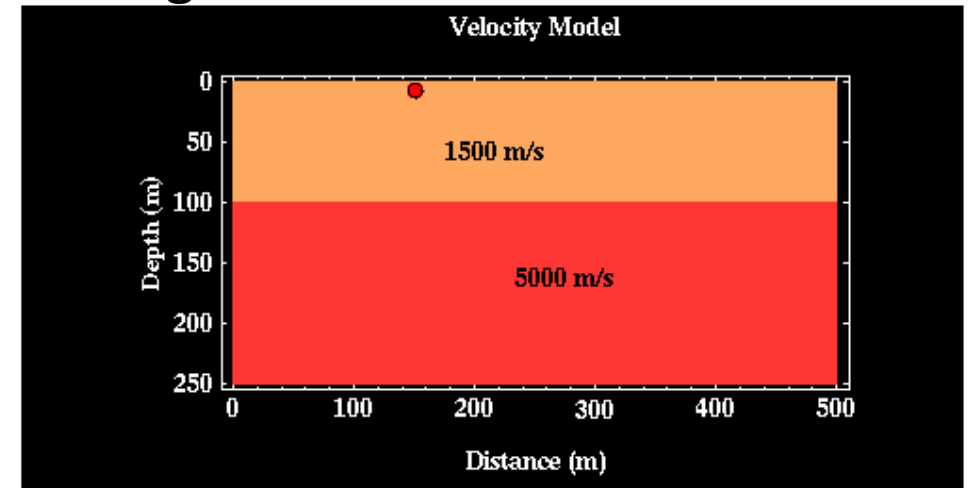
Head waves and the critical angle

Snapshots of numerically simulated
wave fronts: 110 ms

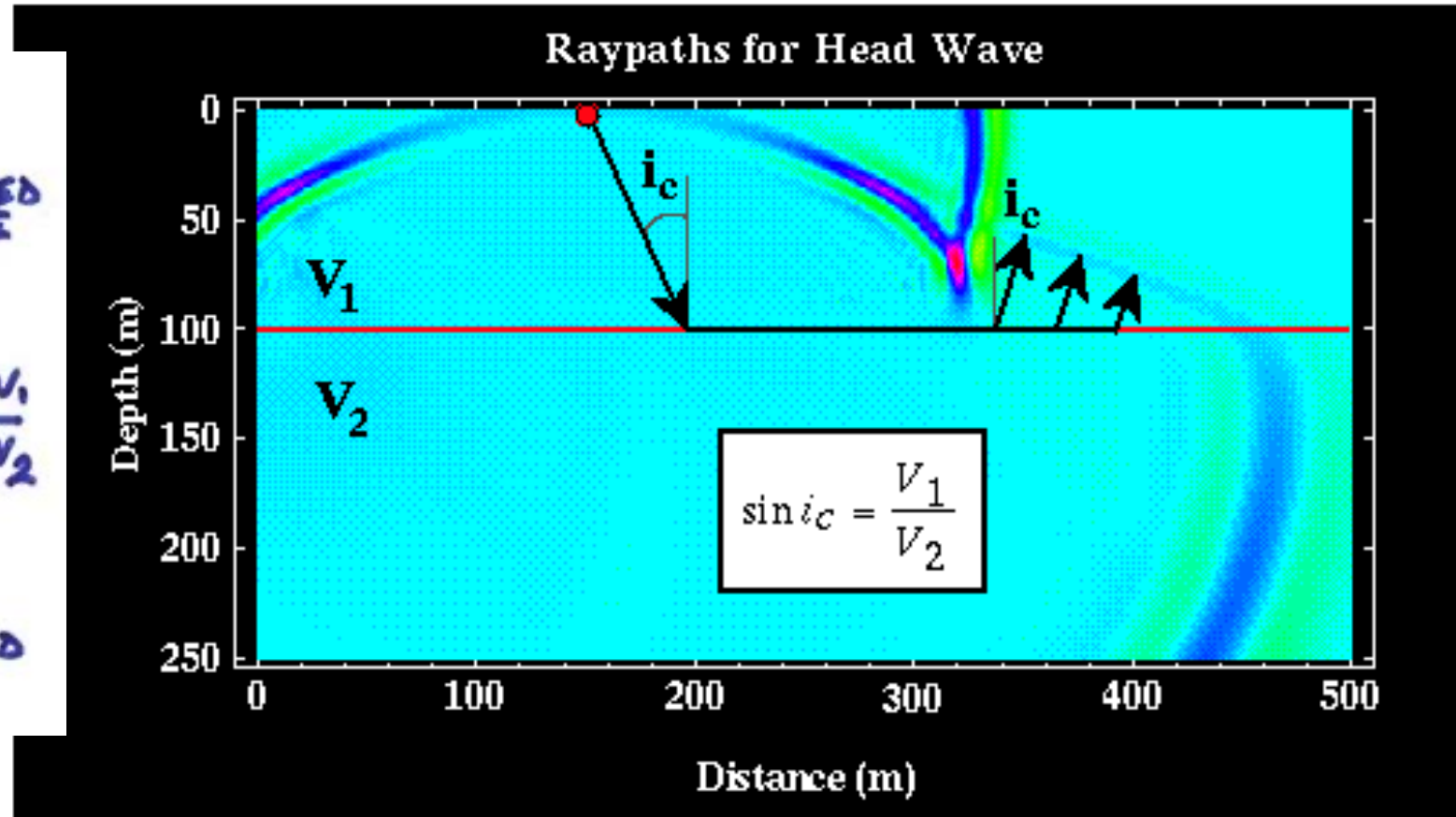
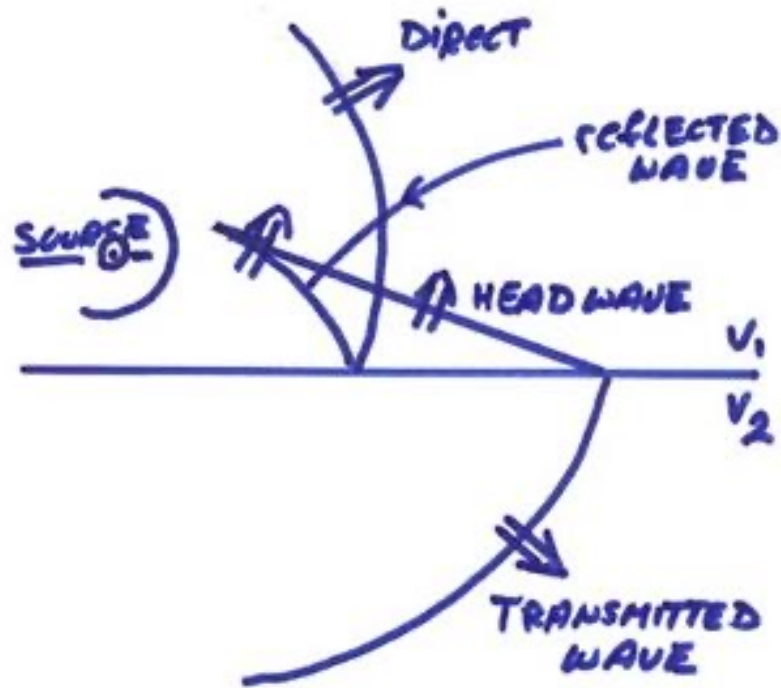


Head waves and the critical angle

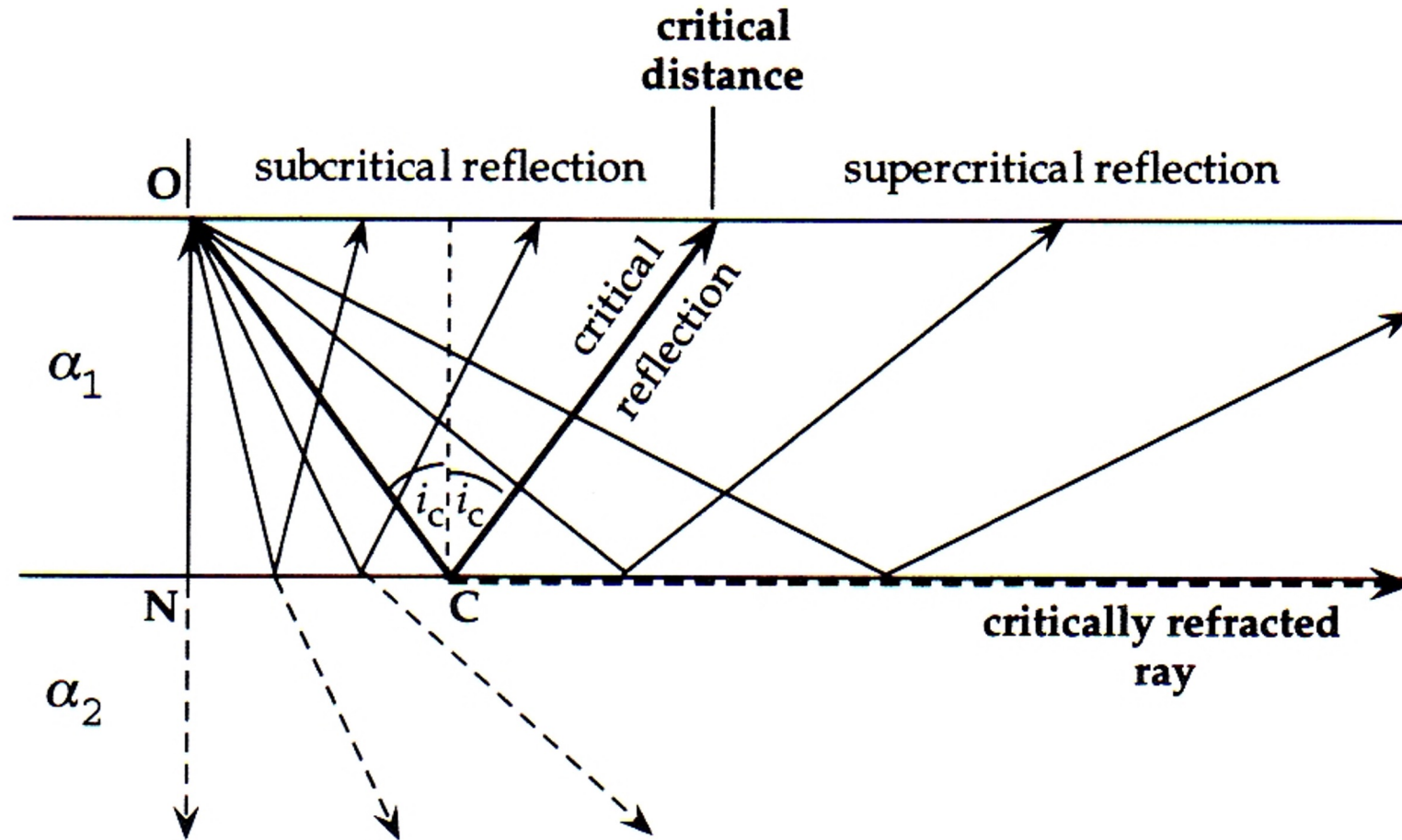
Snapshots of numerically simulated
wave fronts: 140 ms



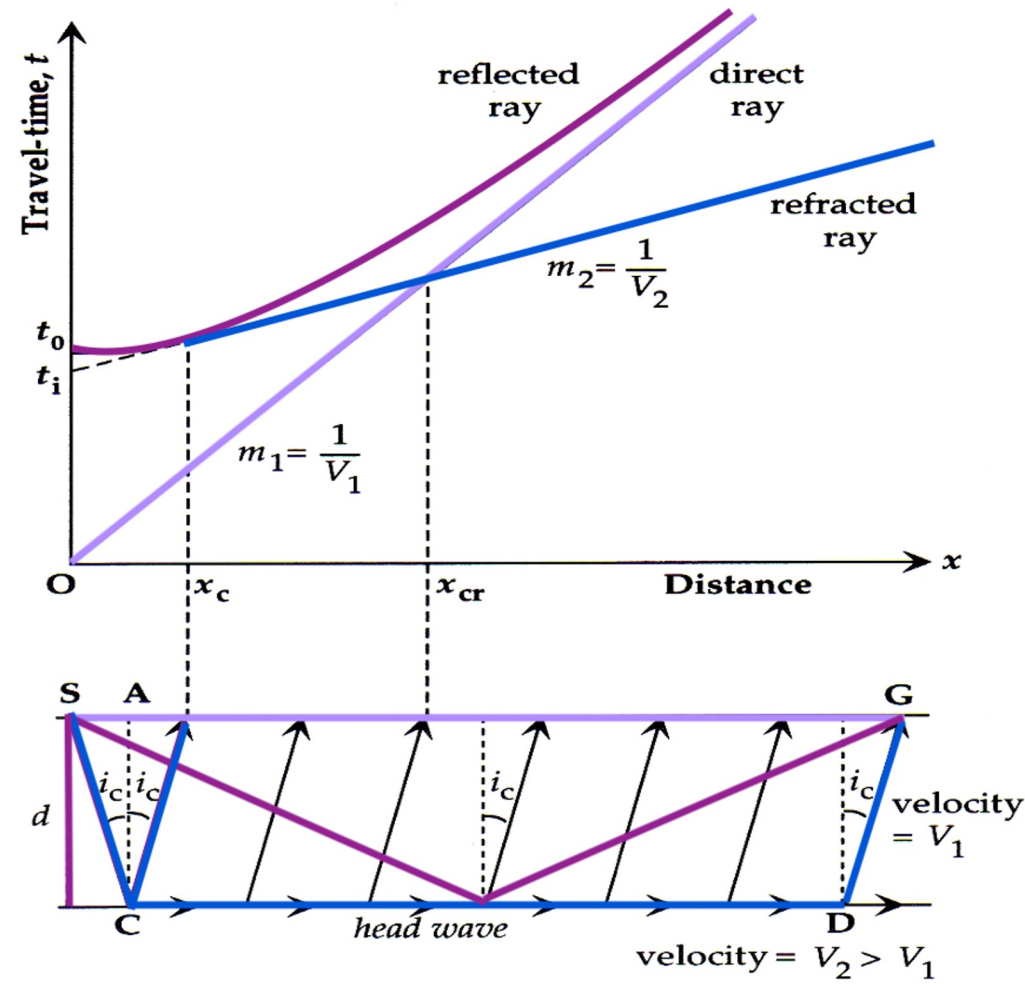
Head waves and the critical angle



The critical angle and the critical distance



Travel time curves



Ray tracing

In ray tracing, we assume that at very small distances, the ray propagation is locally straight. When reaching a new medium, the wave propagates to a new direction.

In the pictures, a ray is passing through a medium with decreasing index of refraction (meaning, the velocity increases with depth).

In panel b, the wave is presented on a large scale.

